# CLEMENT HERITIER

## Embedded Systems & Robotics Engineer

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Embedded Systems and Robotics Engineer with a double MSc (Heriot-Watt / ESIEE Paris) and **2+ years** across rail, automotive and power electronics. Build validation and diagnostic tooling end-to-end: Robot Framework automation, Python/API backends, MATLAB/Simulink model-based design, STM32 and PCB bring-up. Seeking **Embedded Systems and Robotics Engineer role** in **Paris, FR** from **Jan 2027**.

## EXPERIENCE

**Lab Test Automation Engineer – VIE**
Alstom
Jun 2025 – Present
Pittsburgh, US

- Designed and built a full-stack network analysis platform (real-time data packet capture, visualisation, diagnostics).
- Automated end-to-end validation in virtualised environments with Robot Framework.
- Delivered API-driven backend services and GUI tooling that reduced system-level debug time.

**Model-Based & Function Development Engineer – Internship**
Bosch
Sep 2024 – Mar 2025
Paris, FR

- Deployed MATLAB/Simulink model-based design workflows across **2** distributed teams, compliant with automotive (ISO 26262) standards.
- Raised MCDC model test coverage from **81%** to **98%** using Simulink Design Verifier.
- Diagnosed and fixed execution and compilation failures in Processor-in-the-Loop (PIL) testing on embedded targets.

**Embedded Systems Engineer – Internship**
Andrice
May 2023 – Aug 2023
Lyon, FR

- Designed a **500 W** isolated Phase-Shift Full-Bridge from LTSpice modeling and component sizing to bench validation.
- Delivered the full schematic and PCB layout in Altium Designer, then assembled and tested the prototype with STM32-generated PWM and oscilloscope measurements.

## PROJECTS

**Path Planning for Robotic Surface Vehicles – MSc Thesis**
Heriot-Watt University
May 2024 – Aug 2024
Edinburgh, UK

- Benchmarked **6** planners (A*, Dijkstra, BFS, PRM, RRT*, Hybrid A*/Theta*) for autonomous waste-collection USVs, scored on execution time, iterations, path smoothness, clearance and collision-free validity via MATLAB.
- Modelled the BlueBoat hull in MATLAB with the MSS marine-systems toolbox and quantified the planning/actuation trade-off under a **0.3 m/s** current: A* planned fastest (**47 ms**, 44 iterations) but Hybrid A*/Theta* cut power draw **8.9** → **6.7 W/h** and heave settling time **22 s** → **8 s**.

**"Voix de l'Univers" – EM Signal Detection**
ESIEE Paris
Apr 2022 – Jun 2022
Paris, FR

- Built two receivers for lightning and solar-flare emissions: a ferrite-rod antenna (**500 Hz–100 kHz**) and a **9-photodiode** hemispherical array.
- Acquired signals through a PC sound card and analysed spectra by FFT in Spectrum Lab and Audacity.

## EDUCATION

**MSc Robotics**
Heriot-Watt University
Sep 2023 – Sep 2024 • Edinburgh, UK
Double-degree programme with ESIEE Paris.

**MSc Embedded & Electrical Systems**
ESIEE Paris
Sep 2021 – Sep 2024 • Paris, FR

**CPGE Mathematics & Physics**
Lycée Camille Vernet
Sep 2018 – Sep 2021 • Valence, FR

**Scientific Baccalaureate**
Institution Notre Dame
Sep 2018 • Valence, FR

## SKILLS

**PROGRAMMING**
Python | C/C++ | VHDL | Java | HTML/CSS

**EMBEDDED & ELECTRONICS**
STM32 | ESP32/ESP8266 | Altium Designer | LTSpice

**SOFTWARE & TOOLS**
Robot Framework | MATLAB/Simulink | ROS | Docker | Git/GitHub | IBM Rhapsody

**LANGUAGES**
French – Native | English – Fluent | Spanish – Basic

## CERTIFICATIONS

- TOEIC (2023) — **905/990**
- Certification Voltaire (French)

## INTERESTS

- Competitive rugby (16 years)
- Robotics & aerospace
- International travel